NGHIA VAN VO | Curriculum Vitae

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Education

OAKLAND UNIVERSITY
Doctor of Philosophy in Applied Mathematics — 2022–Present, Rochester, Michigan, USA

OAKLAND UNIVERSITY
Master of Science in Applied Statistics — 2022–2024, Rochester, Michigan, USA

UNIVERSITY OF EDUCATION
Bachelor of Science in Applied Mathematics — 2017–2021, Ho Chi Minh City, Vietnam

RESEARCH INTEREST
Convex and Nonconvex Optimization; Nonlinear Optimization; Variational Analysis, Mathematical Programming, Stability Theory, Stochastic Optimization Algorithms.

Publications

PUBLISHED & SUBMITTED MANUSCRIPT

1. (with Tran T. A. Nghia and Khoa V. H. Vu) Nonsmooth Newton methods with effective subspaces for polyhedral regularization, to appear in **Mathematics of Operations Research** (accepted 09/2026).
2. (with Tran T. A. Nghia and Huy N. Pham) Stable recovery of regularized linear inverse problems, **Inverse Problems** 41(6), 065018 (2025).
3. (with Cao-Kha Doan and Gia-Bao Nguyen) A regularity result via fractional maximal operators for p-Laplace equations in weighted Lorentz spaces, **Complex Variables and Elliptic Equations** 67(7), 1737–1755 (2022).
4. (with Thanh-Nhan Nguyen, Minh-Phuong Tran, and Cao-Kha Doan) A gradient estimate related fractional maximal operators for a p-Laplace problem in Morrey spaces, **Taiwanese Journal of Mathematics** 25(4), 809–829 (2021).
5. (with Tran T. A. Nghia and Khoa V. H. Vu) A globally convergent Newton method for nonsmooth optimization in the absence of smooth surrogates, submitted to **Journal of Scientific Computing** (2026).
6. (with Tran T. A. Nghia, Huy N. Pham, and Khoa V. H. Vu) Isolated calmness of regularized linear inverse problems, submitted to **Optimization** (2026).
7. (with P. D. Khanh, V. V. H. Khoa, B. S. Mordukhovich, and D. B. Tran) Inexact DC algorithms in Hilbert spaces with applications to PDE-constrained optimization, submitted to **Applied Mathematics & Optimization** (2026).
8. (with Hoang-Chau Luong, Kaiqi Zhao, and Lingwei Chen) Consistently informative soft-label temperature for knowledge distillation, submitted to **International Conference on Artificial Intelligence and Statistics** (2027).

PREPRINTS

9. (with Tran T. A. Nghia and Khoa V. H. Vu) Nonsmooth Newton methods with effective subspaces for generalized equations.
10. (with Tran T. A. Nghia, Samik Barman, and Khoa V. H. Vu) Quadratically convergent Newton methods with generalized effective subspaces for large-scale Group Lasso-regularized optimization problems.
11. (with Tran T. A. Nghia, Huy N. Pham, and Khoa V. H. Vu) Lipschitz stability of regularized linear inverse problems.

Teaching Experience

OAKLAND UNIVERSITY
Workshop Instructor and Tutor — 2022–Present

- Workshop instructor for Elementary Algebra, Intermediate Algebra, and Precalculus, teaching approximately 80 students through guided questioning and student-developed solutions.
- Tutored weekly at Math Help Center for courses such as Algebra, Calculus I, Calculus II and Multivariable Calculus.
- Graded assignments for Mathematical Analysis for Engineers I and II and Introduction to Differential Equations with Matrix Algebra.

STEAM FOR VIETNAM FOUNDATION — Summer 2023

Teaching Fellow - Introduction to Computational Thinking and Scratch Programming

- Collaborated with program and human-resources teams to deliver a train-the-trainer program for approximately 4,000 Vietnamese teachers on Scratch programming language.
- Trained and mentored nearly 60 teaching assistants in instructional tools and Scratch projects.
- Developed curriculum and training materials as a member of the technical team.

THE ENGINEERING SOCIETY OF DETROIT — Summer 2024

Mathematics Teacher - Boys in Engineering Academy (BEA)

- Facilitated hands-on, project-based mathematics and STEM activities for Detroit middle school students.
- Mentored small groups and provided individualized support during engineering activities.
- Helped students connect quantitative reasoning with engineering design concepts.

Honors & Awards

1. McKay Outstanding Graduate Research Fellowship, Oakland University (2026)
2. Travel Grant to attend the Midwest Optimization Meeting (MoM), South Dakota State University (2026)
3. Travel Grant to attend the Midwest Optimization Meeting (MoM), University of North Dakota (2025)
4. Travel Grant to attend the East Coast Optimization Meeting (ECOM), George Mason University (2025)
5. Travel Grant to attend the Midwest Optimization Meeting (MoM), University of Michigan (2023)
6. Travel Grant to attend the International Conference on Optimization and Variational Analysis with Applications, Vietnam Institute for Advanced Study in Mathematics, Ha Noi, Vietnam (2023)
7. The Award for Teaching Excellence Internship, Department of Mathematics, Binh Phu High School (2021)
8. First Prize at Department of Mathematics Undergraduate Scientific Research Competition (2020)
9. First Prize at Undergraduate Scientific Research Competition, H.C.M City University of Education (2020)

Talks

1. *A Globally Convergent Newton Method for Nonsmooth Optimization in the Absence of Smooth Surrogates*, SIAM DC–Maryland–Virginia Sectional Meeting, George Mason University (2026).
2. *Nonsmooth Newton Methods with Effective Subspaces for Polyhedral Regularization*, Great Lakes SIAM Annual Meeting, Purdue University (2026).
3. *Effective-Subspace Second-Order Methods for Nonsmooth Composite Optimization with Polyhedral Regularizers*, the 28th Midwest Optimization Meeting, South Dakota State University (2026).
4. *Fast Second-Order Methods in Nonsmooth Composite Convex Optimization with Applications*, Graduate Student Research Conference, Oakland University (2026).
5. *Stability Theory of Regularized Linear Inverse Problems through the Lens of Second-Order Variational Analysis*, the 27th Midwest Optimization Meeting, University of North Dakota (2025).
6. *Stable Recovery of Regularized Linear Inverse Problems*, the 5th East Coast Optimization Meeting, George Mason University (2025).
7. *An Introduction to Optimization for Big Data: Data Analysis and Wrangling*, Data Analytics in Multi-Omics Science (DAMOS) Workshop, Oakland University (2026).

Skills

Programming & Development	Python, MATLAB, SAS, C, and SQL; Git and GitHub.
Optimization Algorithms	Gradient and stochastic gradient methods; proximal methods and FISTA; Newton methods; coordinate descent; ADMM; Douglas–Rachford splitting; and the Chambolle–Pock primal–dual method.
Scientific Computing & ML	NumPy, Pandas, Scikit-learn, PyTorch, Matplotlib, and Seaborn.
Optimization Software	CVXPY, Gurobi, CPLEX, and MATLAB Optimization Toolbox.
Certificates	*Machine Learning Specialization*, Stanford University and Coursera; *Mathematics for Machine Learning*, Imperial College London and Coursera; *Scientific Computing and Data Analysis with Python*, freeCodeCamp; *Introduction to Data Analytics in Google Cloud*.
Languages	English and Vietnamese.

References

Research References

Nghia Tran (Supervisor)

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1114 Burnside Hall, 805 Sherbrooke St. W., Montréal, Québec H3A 0B9, Canada
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Boris S. Mordukhovich

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Teaching References

Gerald Kotasek

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Janice Moore

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